

Project Overview Document

Employee Management System

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System Explanation

System Components:

1. Login Page
2. Employee Management
3. Attendance System
4. LEAVE MANAGEMENT SYSTEM
5. Grievance System
6. Dashboard

1.Login Page (Admin/Emp Page)

Login (Admin/Emp Page)

- Email
- Password
- Forgot pass Link

Redirect:

- Admin → Admin Dashboard
- Employee → Employee Dashboard

2.EMPLOYEE MANAGEMENT SYSTEM (Admin Page)

Employee List Page

- Table:
 - ID
 - Name
 - Department
 - Status (Active/Inactive)

- Search bar
- Filter (department, status)
- Add Employee button

Add Employee Form (Admin Page)

- Full Name
- Email
- Phone
- Position
- Department
- Hire Date
- Role (Admin / Employee)
- Address
 - Country
 - City/state

Employee Detail Page (Admin/Emp Page)

- Basic Info
- Attendance summary

Edit / Delete Employee (Admin Page)

- Update info
- Soft delete

3. ATTENDANCE SYSTEM

Attendance Marking (Employee Page)

This is where employees interact with the system daily.

- Check-in/Check-out Card:
 - **Digital Clock** (Current Time).
 - **Action Button:** Large toggle button for "Check-in" or "Check-out."
 - **(Optional) QR Scanner:** Button to launch camera for scanning a workplace QR code.
 - **Today's Status:** A small summary showing the time logged for the current day.

- **Personal Logs:** A table showing the employee's own history for the current month.

Attendance Management (Admin Page)

This provides the oversight and configuration tools for HR/Management.

- **Master Records Table:**
 - **Columns:** ID, Employee Name, Date, Check-in Time, Check-out Time, Total Hours, Status (Present, Late, Absent).
 - **Search/Filter:** Search by name; filter by date range or status.
- **System Controls:**
 - **Toggle Switch:** Enable/Disable Attendance Marking (useful for holidays or system maintenance).
 - **Import Tool:** "Upload Excel" button to bulk-upload records from external biometric devices.
- **Reports & Export:**
 - Button to generate and download monthly PDF/CSV attendance summaries.

Attendance Logic & Data Points

These are the backend fields and rules the system will track:

- **Status Indicators:**
 - **Present:** Checked in before the grace period.
 - **Late:** Checked in after a specific set time (e.g., 9:15 AM).
 - **Absent:** No record found for a working day.
- **Required Fields:**
 - Employee_ID
 - Date
 - Check_in_Time
 - Check_out_Time
 - **Device_Info (Optional):** to track if they used the web portal or QR code).
- **Updated Dashboard Integration**
 - To make the system feel cohesive, these attendance metrics should reflect on your dashboards:
- **Admin Dashboard:**
 - Metric: "Present Today" (Count of employees who checked in).
 - Metric: "Late Arrivals" (Count of employees flagged as Late today).
- **Employee Dashboard:**
 - Metric: "Attendance This Month" (Percentage of days present).
 - Quick Action: "Check-in" button available directly on the main dashboard for speed.

4.LEAVE MANAGEMENT SYSTEM

Leave Request Form (Emp Page)

- Leave type (E.g. Annual, Sick)
- Start date
- End date
- Reason
- Show Leave Balance
 - Annual leave (E.g. 20 total ,15 used ,5 remaining)
- Sick leave (E.g. 10 total ,2 used ,8 remaining)

Leave Request Table (Admin Page)

Table:

- Emp Name
- Leave Type
- Start Type
- End Date
- **Detail (Button):**
 - Emp Name
 - Emp Id
 - Leave type
 - Start Date
 - End Date
 - Reason
 - Remaining Balance
 - Leave History
 - Status (Pending /Approved /Rejected)
 - Approve/Reject (Button)

5.Grievance System

Grievance Form (Emp Page)

- Title
- Description
- Category:
 - Workplace issue
 - Harassment

- Salary
- Other
- Anonymous (checkbox)

Grievance Table (Admin Page)

Table:

- Name (Emp Name/Anonymous)
- Title
- Category
- Detail:
 - Full description
 - Date submitted
 - Status (Under review, Resolved, Rejected)
 - Action:
 - Under review
 - Resolved
 - Rejected

6. Dashboard

Admin Dashboard

- Total Employees
- Present Today
- Pending Leave Requests
- Open Grievances

Employee Dashboard

- Employees Info (Simple)
- Leave balance
- recent requests

BACKEND (NestJS) STRUCTURE

backend/

```

├── src/
│   ├── auth/                                # Login system
│   │   ├── auth.controller.ts
│   │   ├── auth.service.ts
│   │   └── auth.module.ts

```

```
| | └─ dto/
| | └─ guards/
|
| └─ users/                # Employees (core entity)
| | └─ users.controller.ts
| | └─ users.service.ts
| | └─ users.module.ts
| | └─ dto/
| | └─ entities/
|
| └─ attendance/
| | └─ attendance.controller.ts
| | └─ attendance.service.ts
| | └─ attendance.module.ts
| | └─ dto/
| | └─ entities/
|
| └─ leave/
| | └─ leave.controller.ts
| | └─ leave.service.ts
| | └─ leave.module.ts
| | └─ dto/
| | └─ entities/
|
| └─ grievance/
| | └─ grievance.controller.ts
| | └─ grievance.service.ts
| | └─ grievance.module.ts
| | └─ dto/
| | └─ entities/
|
| └─ dashboard/
| | └─ dashboard.controller.ts
| | └─ dashboard.service.ts
```

```

|   |   └─ dashboard.module.ts
|
|   └─ prisma/                  # Prisma setup
|   |   └─ prisma.service.ts
|   |   └─ prisma.module.ts
|
|   └─ common/                  # shared stuff
|   |   └─ guards/
|   |   └─ interceptors/
|   |   └─ filters/
|   |   └─ decorators/
|
|   └─ main.ts
|   └─ app.module.ts
|
└─ prisma/
  └─ schema.prisma              # DB schema (VERY IMPORTANT)
|
└─ package.json
└─ tsconfig.json

```

FRONTEND (Next.js) STRUCTURE

```

frontend/
└─ app/
  └─ login/
    └─ page.tsx
  |
  └─ dashboard/
    └─ admin/
      └─ page.tsx
    └─ employee/
      └─ page.tsx
  |

```

```

|   |— employees/
|   |   |— page.tsx           # list
|   |   |— add/
|   |       |— page.tsx
|   |       |— [id]/
|   |           |— page.tsx   # detail
|
|   |— attendance/
|   |   |— page.tsx
|   |   |— admin/
|   |       |— page.tsx
|
|   |— leave/
|   |   |— request/
|   |       |— page.tsx
|   |       |— admin/
|   |           |— page.tsx
|
|   |— grievance/
|   |   |— page.tsx
|   |   |— admin/
|   |       |— page.tsx
|
|   |— layout.tsx
|— components/
|   |— ui/
|   |— forms/
|   |— tables/
|   |— cards/
|

```

```

├─ lib/
|   ├─ api.ts          # axios / fetch setup
|   └─ auth.ts
|
├─ styles/
|   └─ globals.css
├─ tailwind.config.js
└─ package.json

```

Database Structure

ENTITIES:

1. Employee
2. Department
3. Address
4. Attendance
5. Leave_Request
6. Leave_Type
7. Leave_Balance
8. Grievance
9. Session / Token (optional)

1.Employee:

Field	Type	Constraints	Notes
employee_id	UUID	PK	
full_name	VARCHAR(100)	NOT NULL	
email	VARCHAR(150)	UNIQUE, NOT NULL	
password	TEXT	NOT NULL	hashed
phone	VARCHAR(20)		optional
role	VARCHAR(20)	NOT NULL	Admin / Employee
status	VARCHAR(20)	DEFAULT 'Active'	Active / Inactive
hire_date	DATE	NOT NULL	
position	VARCHAR(100)		
department_id	UUID	FK → department	

address_id	UUID	FK → address	
created_at	TIMESTAMP	DEFAULT NOW()	
updated_at	TIMESTAMP		

2.Department:

Field	Type	Constraints	Notes
department_id	UUID	PK	
name	VARCHAR(100)	UNIQUE, NOT NULL	
description	TEXT		

3.Address:

Field	Type	Constraints	Notes
address_id	UUID	PK	
country	VARCHAR(100)	NOT NULL	
city	VARCHAR(100)	NOT NULL	
state	VARCHAR(100)		

4.Attendance:

Field	Type	Constraints	Notes
attendance_id	UUID	PK	
employee_id	UUID	FK → employee	
date	DATE	NOT NULL	one per day
check_in_time	TIMESTAMP		
check_out_time	TIMESTAMP		
total_hours	DECIMAL(5,2)		calculated
status	VARCHAR(20)	NOT NULL	Present / Late / Absent
device_info	VARCHAR(100)		optional

5. Leave_Request:

Field	Type	Constraints	Notes
leave_id	UUID	PK	
employee_id	UUID	FK → employee	
leave_type_id	UUID	FK → leave_type	
start_date	DATE	NOT NULL	
end_date	DATE	NOT NULL	
reason	TEXT		
status	VARCHAR(20)	DEFAULT 'Pending'	
approved_by	UUID	FK → employee	Admin
created_at	TIMESTAMP	DEFAULT NOW()	

6. Leave_Type:

Field	Type	Constraints	Notes
leave_type_id	UUID	PK	
name	VARCHAR(50)	UNIQUE, NOT NULL	
description	TEXT		
is_paid	BOOLEAN	DEFAULT TRUE	

7. Leave_Balance:

Field	Type	Constraints	Notes
balance_id	UUID	PK	
employee_id	UUID	FK → employee	
leave_type_id	UUID	FK → leave_type	
total_days	INT	NOT NULL	
used_days	INT	DEFAULT 0	
remaining_days	INT	NOT NULL	

8. Grievance:

Field	Type	Constraints	Notes
grievance_id	UUID	PK	

employee_id	UUID	FK → employee	NULL if anonymous
title	VARCHAR(150)	NOT NULL	
description	TEXT	NOT NULL	
category	VARCHAR(50)	NOT NULL	
is_anonymous	BOOLEAN	DEFAULT FALSE	
status	VARCHAR(30)	DEFAULT 'Under Review'	
created_at	TIMESTAMP	DEFAULT NOW()	

9.Session / Token (optional):

Field	Type	Constraints	Notes
token_id	UUID	PK	
employee_id	UUID	FK → employee	
token	TEXT	NOT NULL	JWT or refresh token
expiry_date	TIMESTAMP	NOT NULL	

10.RELATIONSHIP SUMMARY:

From	To	Type
employee	department	Many → One
employee	address	Many → One
employee	attendance	One → Many
employee	leave_request	One → Many
employee	grievance	One → Many
leave_request	leave_type	Many → One
leave_balance	leave_type	Many → One

Backlog

Epics:

1. Authentication System
2. Employee Management

3. Attendance System
4. Leave Management
5. Grievance System
6. Dashboard & Analytics

1. EPIC: Authentication System

Task [US-1: Login]:

- **Backend:** Create auth module
- **Backend:** Login API (JWT)
- **Backend:** Password hashing
- **Frontend:** Login UI
- **Frontend:** Form validation
- **Frontend:** Store token
- **Frontend:** Redirect based on role

Task [US-2: Forgot Password]:

- **Backend:** Generate reset token
- **Backend:** Email service (or mock)
- **Frontend:** Forgot password page
- **Frontend:** Reset password form

2. EPIC: Employee Management

Task [US-3: View Employee List (Admin)]:

- **Backend:** GET /employees
- **Backend:** Filtering (department, status)
- **Frontend:** Table UI
- **Frontend:** Search input
- **Frontend:** Filter dropdowns

Task [US-4: Add Employee]:

- **Backend:** POST /employees
- **Backend:** Validation (email unique)
- **Frontend:** Form UI
- **Frontend:** Submit form
- **Backend:** Assign default leave balance

Task [US-5: Employee Detail Page]:

- **Backend:** GET /employees/:id
- **Backend:** Join attendance + leave
- **Frontend:** Detail page UI

Task [US-6: Edit/Delete Employee]:

- **Backend:** PATCH /employees/:id
- **Backend:** Soft delete logic
- **Frontend:** Edit form
- **Frontend:** Delete button + confirm modal

3.EPIC: Attendance System

Task [US-7: Check-in / Check-out (Employee)]:

- **Backend:** POST /attendance/check-in
- **Backend:** POST /attendance/check-out
- **Backend:** Calculate total hours
- **Backend:** Late logic
- **Frontend:** Clock UI
- **Frontend:** Button toggle

Task [US-8: View Personal Attendance]

- **Backend:** GET /attendance/me
- **Frontend:** Table UI

Task [US-9: Admin Attendance Management]

- **Backend:** GET /attendance (all)
- **Backend:** Filters (date, status)
- **Frontend:** Admin table

Task [US-10: Attendance Reports]

- **Backend:** Generate report (CSV/PDF)
- **Frontend:** Download button

4.EPIC: Leave Management

Task [US-11: Submit Leave Request]

- **Backend:** POST /leave
- **Backend:** Validate balance
- **Frontend:** Form UI

Task [US-12: View Leave Balance]

- **Backend:** GET /leave/balance
- **Frontend:** Display balances

Task [US-13: Admin Approve/Reject Leave]

- **Backend:** PATCH /leave/:id/status
- **Backend:** Deduct balance
- **Frontend:** Admin table

- **Frontend:** Approve/Reject buttons

Task [US-14: Leave History]

- **Backend:** GET /leave/history
- **Frontend:** Table UI

5.EPIC: Grievance System

Task [US-15: Submit Grievance]

- **Backend:** POST /grievance
- **Frontend:** Form UI
- **Frontend:** Anonymous checkbox

Task [US-16: View Grievances (Admin)]

- **Backend:** GET /grievance
- **Frontend:** Table UI

Task [US-17: Update Grievance Status]

- **Backend:** PATCH /grievance/:id
- **Frontend:** Status buttons

6.EPIC: Dashboard

Task [US-18: Admin Dashboard]

- **Backend:** GET /dashboard/admin
- **Metrics:**
 - Total employees
 - Present today
 - Pending leaves
 - Open grievances
- **Frontend:** Cards UI

Task [US-19: Employee Dashboard]

- **Backend:** GET /dashboard/employee
- **Frontend:** Cards + quick actions

Leave Rules in Ethiopia

Paid Leave

- Annual

- Sick
- Maternity
- Paternity
- Special (Marriage, Bereavement)
- Public Holidays

Unpaid Leave

- Personal leave
- Extended leave

Conditional Leave

- Union leaves
- Study leave
- Religious leave

1. Annual Leave (Vacation Leave)

- Minimum **16 working days per year** after 1 year of service
- Increases with experience (**+1 day every 2 years**)
- Fully **paid**

2. Sick Leave

- Up to **6 months per year**
- Payment structure:
 - 1st month → **100% salary**
 - 2nd-3rd months → **50% salary**
 - Remaining → **unpaid**
- Requires **medical certificate**

3. Maternity Leave

- Total **120 days (paid)**
 - 30 days before birth
 - 90 days after birth

4. Paternity Leave

- 3 working days (paid) for fathers

5. Special Leave (Family / Personal Events)

Usually **paid short leaves** for:

- Marriage → ~3 days
- Bereavement (death of close family) → ~3 days
- Other serious family events

6. Public Holidays

- Paid leave on officially recognized national holidays

7. Unpaid Leave / Exceptional Leave

- Given for **special circumstances** (e.g., personal issues)
- Usually based on company policy

8. Union Leave / Civic Duty Leave

- For:
 - Trade union activities
 - Court duty / public service
- Usually **paid** depending on situation

9. Other (Optional / Company-Based)

Not strictly required by law but sometimes included:

- Study leave
- Compassionate leave extension
- Religious leave (e.g., pilgrimage)